2017 MJC H2 Chemistry Prelim Paper 2

1 Concepts of acids and bases are important aspects in Chemistry which can be applied to many areas in our daily life.

(a) There are three theories of acids and bases. In 1887, Svante Arrhenius proposed that acids are substances that produce H^+ in water and bases are those which produce OH^- in water. In 1923, Johannes N Brønsted and Thomas M Lowry proposed another set of definitions for acids and bases. Later in the same year, Gilbert N Lewis defined acids and bases in terms of electron pair acceptors and donors respectively.

(i) State the *Bronsted–Lowry theory* of acids and bases.

[1]

(ii) There are several limitations to Arrhenius' theory of acids and bases. Give an example to illustrate one of its limitations.

[1]

(b) The pK_a values and solubilities of some weak acids in methanol and water at 25 ˚C are given in **Table 1.1**.

Table 1.1

compound	pKa in methanol	pKa in water	solubility/ per 100 g of water
CH_3COOH ethanoic acid	9.71	4.88	100 g
$CH_2(CN)COOH$ 2–cyanoethanoic acid	7.50	2.46	More than 100 g

(i) Two samples of 2–cyanoethanoic acid of identical concentrations were dissolved in water and methanol separately. Calculate the ratio of H^+ ions formed in water and in methanol.

[1]

(ii) Suggest a reason why both acids in **Table 1.1** have lower pK_a in water than in methanol.

[1]

(iii) Explain why 2–cyanoethanoic acid has a higher solubility in water than ethanoic acid.

[1]

(c) The base dissociation constants of some weak bases are given in **Table 1.2**.

Table 1.2

compound	K_b value at 25˚C
methylamine CH_3NH_2	4.6×10^{-4}
dimethylamine $(CH_3)_2NH$	5.4×10^{-4}
trimethylamine $(CH_3)_3N$	6.3×10^{-5}

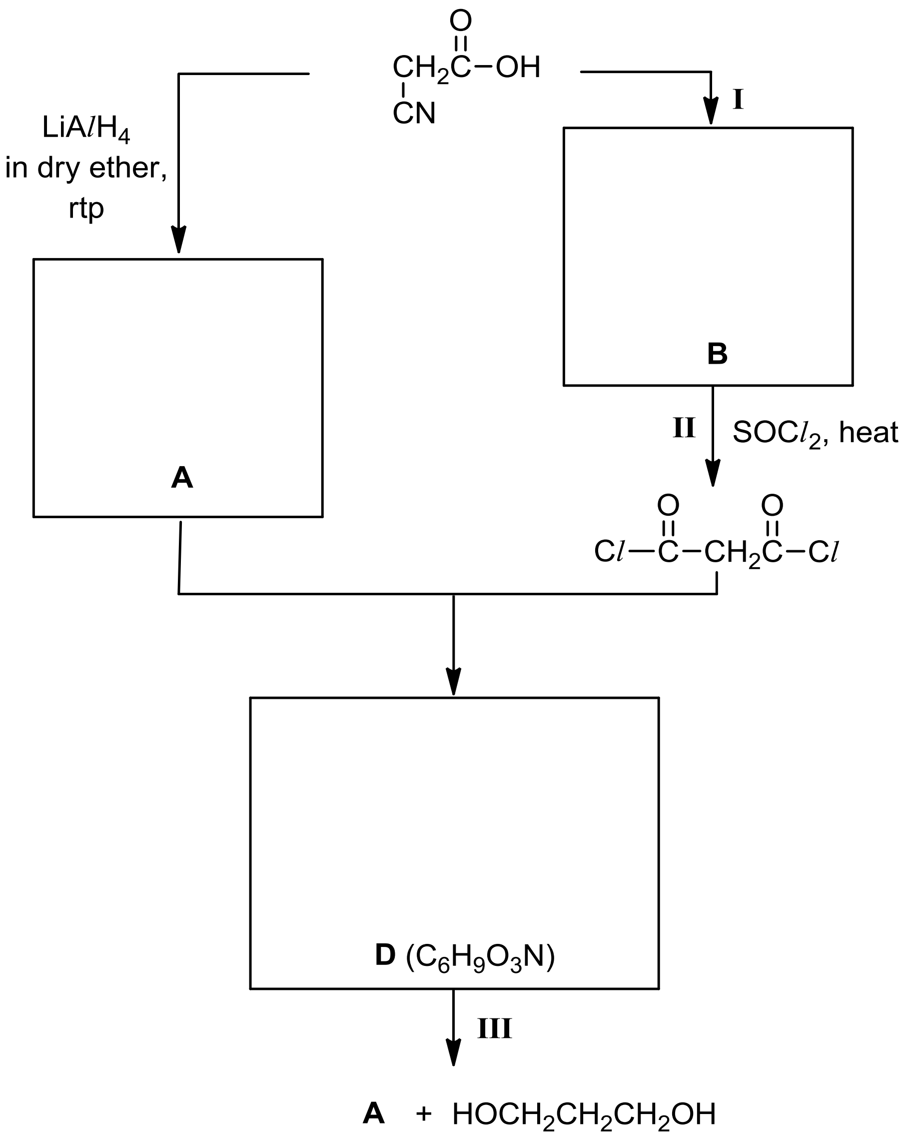
(i) Explain why dimethylamine is more basic than methylamine.

[1]

(ii) Among the three compounds, trimethylamine is the least basic, which is unusual. Suggest a reason for this.

[1]

(d) 2–cyanoethanoic acid can undergo a series of reactions to form a neutral cyclic compound **D**. Compound **D** can further react to produce compound **A** and a diol**.**



(i) Draw the structures of compounds **A**, **B** and **D** in the boxes above. [3]

(ii) State the types of reaction in step **I** and **II.** [1]

Step **I**: ……………………………………….

Step **II**: ……………………………………….

(iii) State the reagents and conditions required in Step **I** and **III**. [2]

Step **I**: ……………………………………….

Step **III**: ……………………………………..

[Total: 13]

2 Vitamins and minerals are essential nutrients that perform many roles in the body. They help to build bones, heal wounds, bolster the immune system and convert food into energy. Young children require many essential minerals such as calcium, magnesium, iodine, iron and zinc to develop and grow.

Table 2.1 shows the recommended daily intake of some essential minerals for children.

Table 2.1

Mineral	**Recommended daily intake for children / mg**	
	Age group	
	1 – 3 years	**4 – 8 years**
Calcium	500	700
Iodine	0.090	0.090
Iron	9	10
Magnesium	80	130
Phosphorus	460	500
Zinc	3	4

Table 2.2 shows part of a nutrition label on a tin of powdered milk formula.

Table 2.2

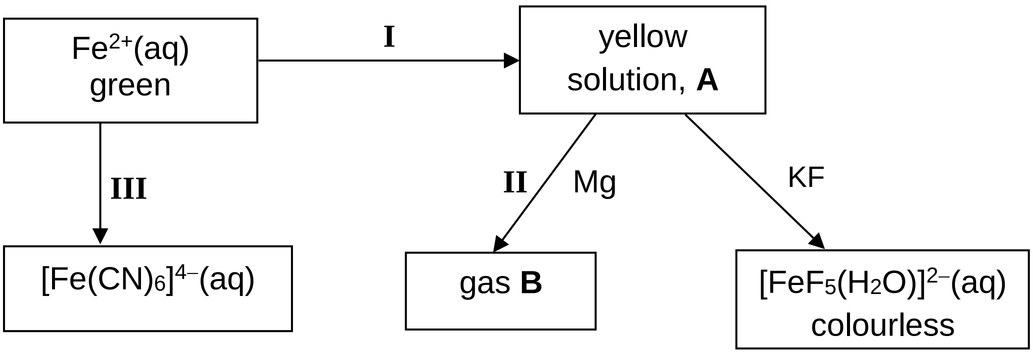
Nutrition Information **Standard Dilution (per 100 ml)**	
Nutrients:	
Protein	2.2 g
Fat	5.1 g
Carbohydrate	11.2 g
Minerals:	
Sodium	39 mg
Potassium	106 mg
Chloride	79 mg
Calcium	119 mg
Phosphorus	69 mg
Magnesium	7.8 mg
Iron	1.03 mg
Zinc	0.7 mg
Copper	0.056 mg
Manganese	0.0094 mg
Iodine	0.011 mg

(a) Zinc helps the immune system fight off invading bacteria and viruses. A 2–year old child takes an average of 3 feeds of milk formula per day, with a quantity of 180 ml per feed.

Using the information provided, determine if the zinc obtained from the milk formula meets the recommended quantity for daily intake. Comment on whether there is a need for the child to supplement his diet with zinc from other sources.

[2]

(b) Iron is an essential mineral used to carry oxygen in the blood.
The following flowchart shows the reactions of iron and its compounds.



(i) Name the types of reactions taking place in **I** and **II**.

[2]

I: ……………………………………….

II: ……………………………………….

(ii) Write the formula of the cation present in **A** and identify gas **B**.
By considering the nature of this cation present, explain fully why gas **B** is formed when Mg is added to **A** in reaction **II**.

[3]

(iii) What is the type of reaction taking place in **III**? Write a balanced equation for the reaction.

[2]

(iv) Fe^{2+}(aq) appears to be green whereas $[FeF_5(H_2O)]^{2-}$(aq) is colourless.

1. Explain why Fe^{2+}(aq) appears to be green.

[3]

2. Write the full electronic configuration of iron in $[FeF_5(H_2O)]^{2-}$.

[1]

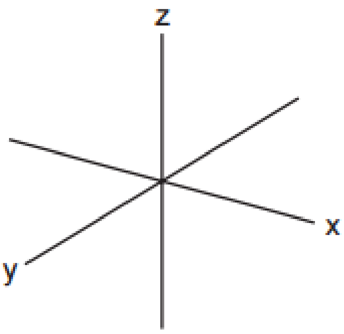
3. Suggest why $[FeF_5(H_2O)]^{2-}$ is colourless.

[1]

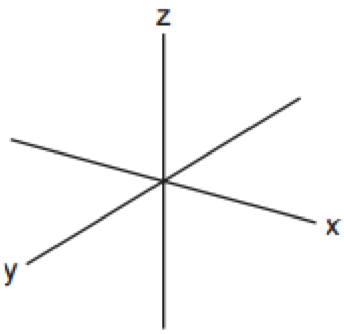
(v) Iron forms many octahedral complexes, of which $[Fe(CN)_6]^{4-}$ is an example. On the axes below, sketch the shapes of a d orbital from the lower energy level and a d orbital from the higher energy level in an octahedral complex.

[2]

lower energy level



higher energy level



(c) Calcium is the most abundant mineral present in milk formula. It is essential for healthy bones and teeth. In the blood, it also helps to regulate the heartbeat, blood pressure and neural transmission.

Table 2.3 gives data about some physical properties of calcium and iron.

Table 2.3

property	calcium	iron
relative atomic mass	40.1	55.8
atomic radius (metallic)/ nm	0.197	0.126
ionic radius (2+)/ nm	0.099	0.076
melting point / °C	839	1538
density / g cm^{-3}	1.54	7.86

(i) Explain why the atomic radius of iron is less than that of calcium.

[2]

(ii) Use relevant data from **Table 2.3** to explain qualitatively why the density of iron is significantly greater than that of calcium.

[1]

(d) Magnesium, phosphorus and chlorides are also minerals found in milk formula.

(i) Write relevant chemical equations to illustrate the reactions of magnesium chloride and phosphorus pentachloride with water. Suggest the approximate pH of any solutions formed.

[2]

(ii) Like magnesium, phosphorus and chlorine, element **Z** is also a Period 3 element. The first six ionisation energies of **Z** are shown in **Figure 2.1** while the 2nd ionisation energies of seven consecutive elements **A – G** (one of which is **Z**) is shown in **Figure 2.2**.

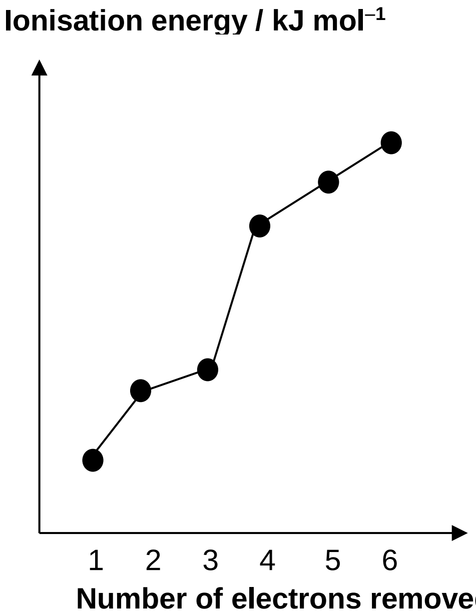


Figure 2.1

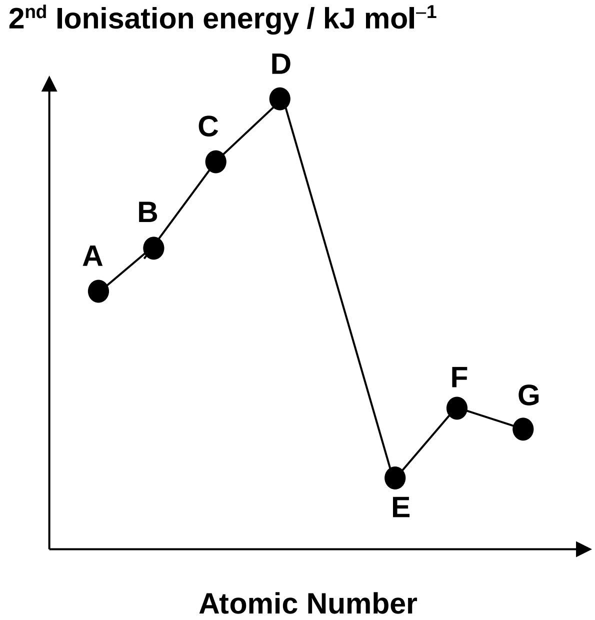


Figure 2.2

Identify element **Z**. Hence, state which element **A – G** in **Figure 2.2**, could be **Z**.

[1]

[Total: 22]

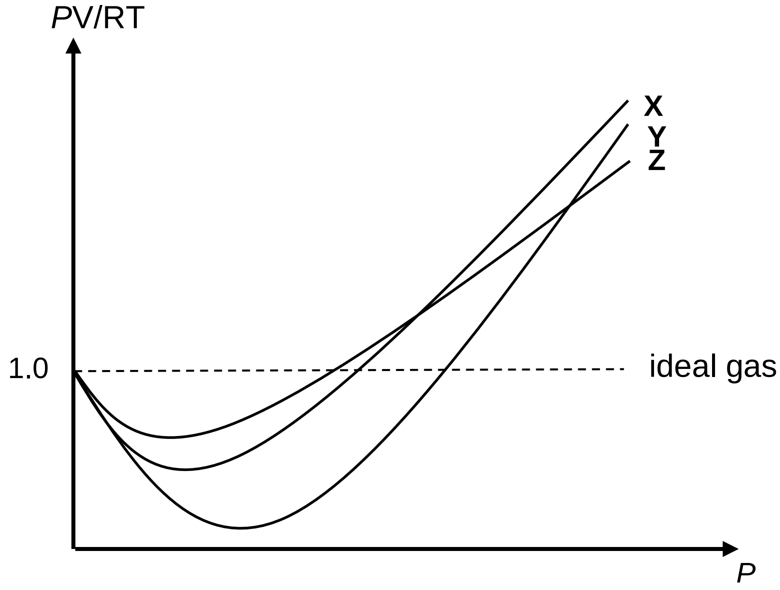
3 Nitrosyl chloride, NOC*l*, is a toxic gas that has caused several industrial accidents.

At 250 °C, NOC*l* readily dissociates into NO and C*l*$_2$.

$$2NOCl\ (g) \rightleftharpoons 2NO\ (g) + Cl_2\ (g)$$

(a) The graphs of *P*V/RT against *P* for 1 mole each of NO, C*l*$_2$ and NOC*l* at room temperature are plotted below.

Figure 3.1



With reference to **Figure 3.1**, deduce which of the graphs is the plot for NOC*l*.

[2]

(b) The dissociation was investigated by adding 5 moles of NOC*l* into a 5.0 dm^3 sealed vessel, and equilibrium was established at 250 °C under a pressure of 5.80 x 10^6 Pa.

(i) Assuming ideal gas behaviour, determine the total amount of gas in moles, at equilibrium.

[1]

(ii) Write the expression for the equilibrium constant, K_p, for the dissociation of NOC*l*, including units.

[1]

(iii) Calculate a value for K_p under the stated conditions.

[3]

(c) Standard Gibbs free energy change, $\Delta G^{\ominus}$ is related to K_c by the following equation.

$$\Delta G^{\ominus} = -RT \ln K_c$$

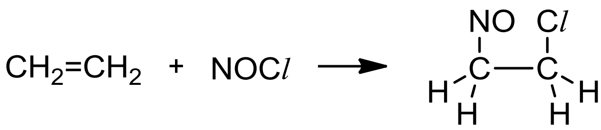
$\Delta G^{\ominus}$ for the dissociation of NOC*l* at 250 °C is –1.47 kJ mol^{-1}.

(i) Using the above information, calculate K_c for the dissociation of NOC*l* at 250 °C. [1]

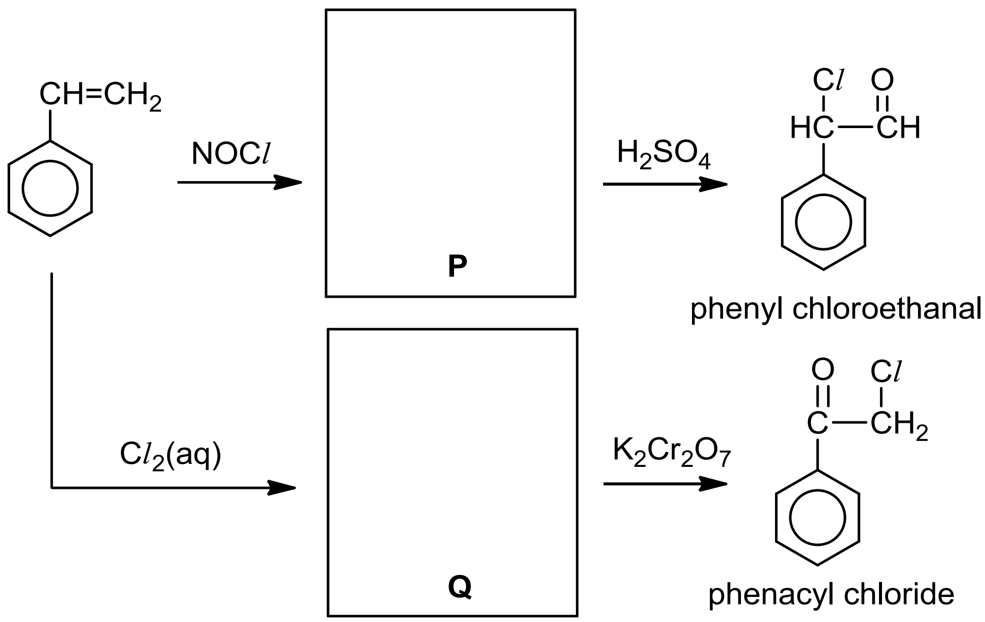
(ii) Suggest how K_c will change when the temperature is raised. [1]

(iii) Hence, using *Le Chatelier's Principle*, deduce whether the dissociation reaction is exothermic or endothermic. [2]

(d) NOC*l* is sometimes used in organic synthesis. It can react with alkenes to form chloro–nitroso compounds according to the following equation.



A reaction scheme involving the use of NOC*l* for the synthesis of phenyl chloroethanal and phenacyl chloride is given below



(i) Draw the structures of compounds **P** and **Q**.

[2]

(ii) State a reagent which can distinguish between phenyl chloroethanal and phenacyl chloride. Write a balanced equation for the reaction.

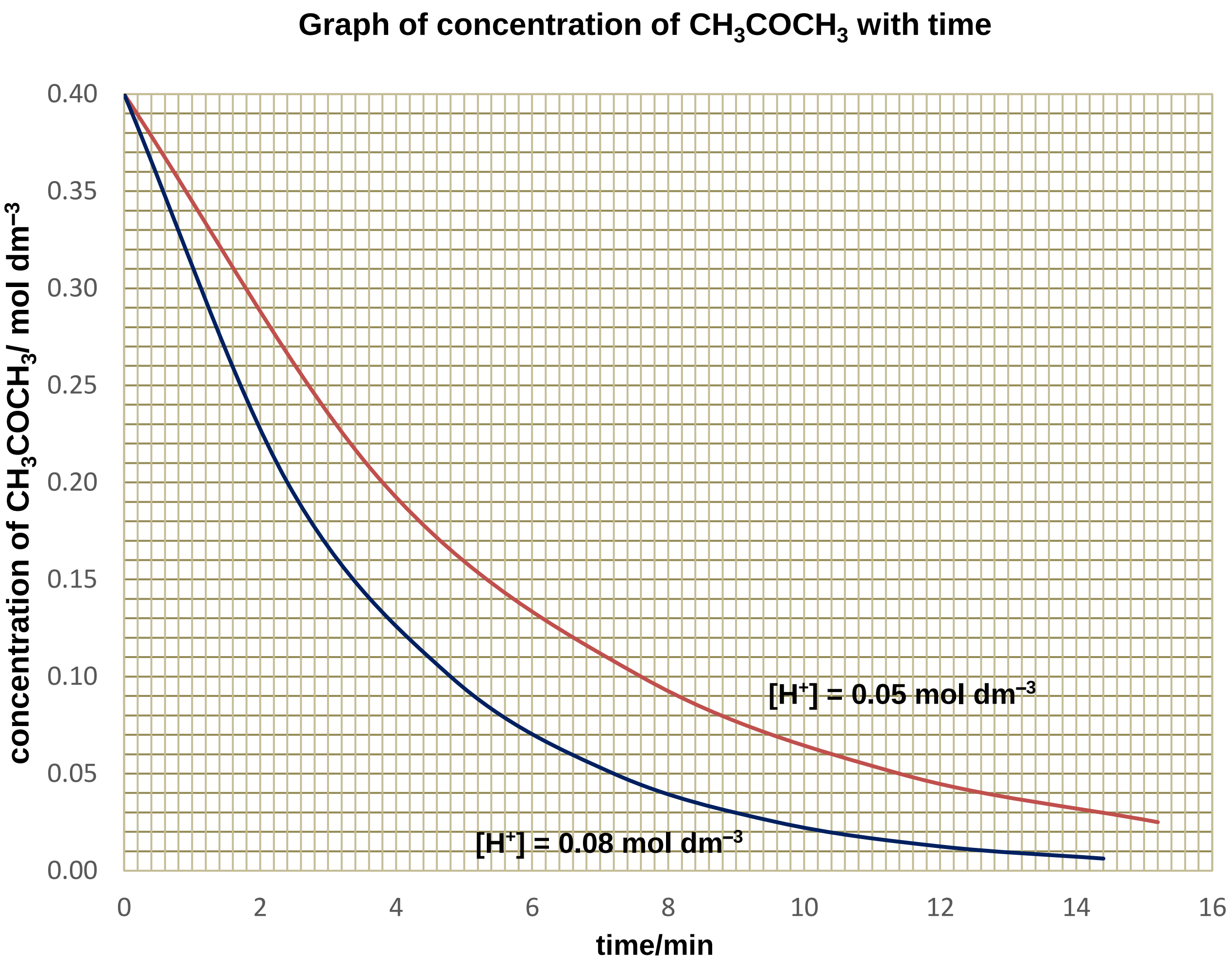
[2]

[Total: 15]

4 The bromination of propanone is an acid–catalysed reaction which produces bromopropanone. The equation for the reaction is given below.

$$CH_3COCH_3 + Br_2 \xrightarrow{H^+} CH_2BrCOCH_3 + H^+ + Br^-$$

(a) Two experiments on the kinetics of the reaction are conducted with [Br_2] being kept constant. The [CH_3COCH_3] is monitored and the following graphs are obtained.



(i) Using the information above, deduce the order of reaction with respect to CH_3COCH_3 and H^+. Show your working on the graph.

[3]

(ii) Another experiment was conducted with a different $[Br_2]$. It was determined that changing $[Br_2]$ has no effect of the rate of reaction. Sketch a graph to show how $[Br_2]$ varies with time.

[1]

(iii) Using your answers from **(a)(i)** and **(a)(ii)**, construct the rate equation for the bromination of CH_3COCH_3.

[1]

(iv) Given that the initial rate of reaction is 7.6×10^{-5} mol dm^{-3} min^{-1} when $[H^+]$ is 0.05 mol dm^{-3}, calculate the rate constant, *k*, for the reaction. State its units.

[1]

(b) Bromopropanone, $CH_2BrCOCH_3$, can also be synthesised via organic reactions.

(i) Devise a 3–step synthesis to produce $CH_2BrCOCH_3$ from the allyl alcohol, $CH_2{=}CHCH_2OH$.

[3]

(ii) Other than the use of an oxidising agent, suggest a chemical test to determine if the synthesis reaction in **(b)(i)** is complete.

[2]

(c) $CH_2BrCOCH_3$ undergoes hydrolysis with aqueous NaOH to form $CH_2(OH)COCH_3$. The reaction follows a second order kinetics.

(i) Outline the mechanism for this reaction, showing all the charges and using curly arrows to represent the movement of electron pairs.

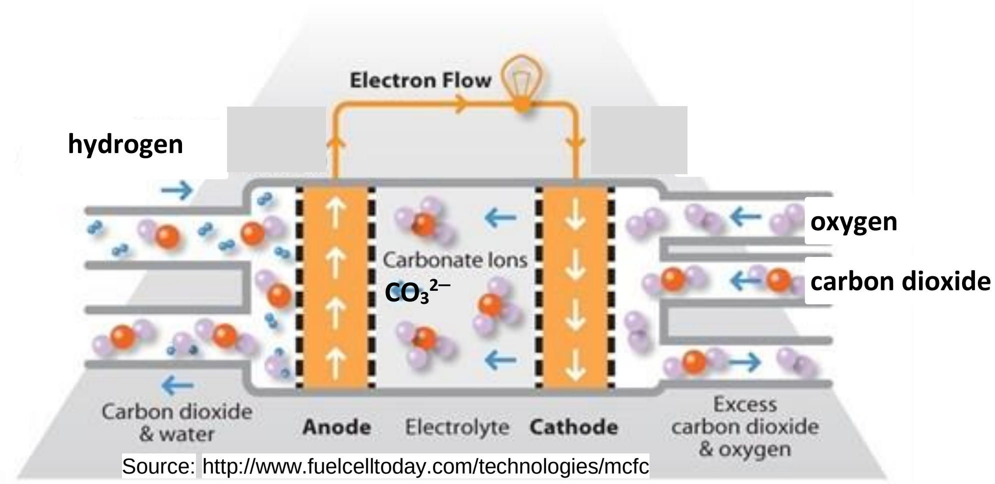
[2]

(ii) CH_3CH_2COBr is an isomer of bromopropanone, $CH_2BrCOCH_3$. The rate of hydrolysis for each of the two compounds is different. State and explain which isomer has a faster rate of hydrolysis.

[2]

[Total: 15]

5 A fuel cell is an electrochemical cell that converts the chemical energy from a fuel into electricity. One example is molten–carbonate fuel cells (MCFCs) developed for natural gas power plants.



(a) At the anode, water and carbon dioxide are formed whereas carbonate ions are formed at the cathode.

(i) Construct the half–equations at the electrodes of this MCFC electrochemical cell. Hence, give the balanced equation for the reaction that occurs during discharge of the MCFC.

[2]

(ii) Draw a fully labelled diagram of the electrochemical cell to determine the standard electrode potential of the O_2(g)/OH^-(aq) electrode system.

[2]

(b) Another example of a fuel cell is direct–formic acid fuel cells (DFAFCs).
It has the overall reaction similar to the combustion of formic acid in oxygen. The following reactions take place at the electrodes.

$CO_2 + 2H^+ + 2e \rightleftharpoons HCOOH \qquad E^\theta = -0.61\ V$

$\frac{1}{2}O_2 + 2H^+ + 2e \rightleftharpoons H_2O \qquad E^\theta = +1.23\ V$

By using one of the phrases *more positive, more negative* or *no change,* deduce the effect of increasing $[OH^-]$ on the electrode potential of the cathode.

[1]

(c) **Table 5.1** shows the different enthalpies of various compounds. The data will be useful for this question.

Table 5.1

	ΔH^θ / kJ mol^{-1}
standard molar enthalpy of formation of CO_2 (g)	−394
standard molar enthalpy of formation of H_2O (*l*)	−286
standard molar enthalpy of formation of CH_3OH (aq)	−246
standard molar enthalpy of formation of HCHO (aq)	−150
standard molar enthalpy of combustion of formic acid, CH_2O_2 (*l*)	−211

(i) Define the term *standard enthalpy change of formation*, ΔH_f^θ of formic acid, CH_2O_2.

[1]

(ii) Using Hess's law and information from **Table 5.1**, construct an energy cycle to calculate the $\Delta H_f^\ominus$ of formic acid.

[3]

(iii) Methanol, CH_3OH is toxic because liver enzymes oxidise it to formaldehyde, HCHO.

Using **Table 5.1**, calculate the standard enthalpy change, $\Delta H^\ominus$ for the following reaction.

$$CH_3OH\ (aq) + \tfrac{1}{2}O_2\ (g) \longrightarrow HCHO\ (aq) + H_2O\ (l)$$

[1]

[Total: 10]

End of Paper 2